L

LCA (*see* Life-Cycle Assessment)

LNG (*see* Liquid Natural Gas)

Laser Cutting, Drilling, Machining, Welding (*see* Metallurgy)

Life-Cycle Assessment (LCA) (of Turbomachinery)*

Life-cycle assessment is the art of being able to use data about:

1. A machine, its operating history, and any failures
2. The fleet of that type and model of machine
3. Changing process conditions in a plant to assess or extend the life of turbomachinery components or the machine itself.

The data can be used to:

1. Avoid catastrophic failures
2. Improve the quality of the machine's operation (lower operational temperature, promoting less hot erosion)
3. Assess better the life cycles being accumulated on a machine to avoid premature overhauls or parts change
4. A combination of all of these

Many consulting companies today sell life-assessment services. Some of them provide a package that includes monitors, interpreting the data for their clients and so forth. Even if clients decide to pay for this service, they should get some basic training for in-house staff, so they can better assess their options. Also, any consulting service could be an interim measure if staff training is provided, so the clients can eventually proceed on their own.

For accurate life-cycle assessment, it is essential that the individual machine's parameters be used. Even among machines of exactly the same type and model, stress levels may be considerably different in actual operation.

Consider the following severe example. In a squadron of aircraft doing acrobatic maneuvers in formation for an air show, the squadron leader, generally out in front, sets the pace. The other aircraft take up their positions, in tight formation, attempting to stay a specific distance from their leader's wing tips. The stress of the aircraft's engines that are not leading is much higher than that of the leader's. This is because their power levels have to "hunt" back and forth to maintain their required positions. These engines may incur stress of as much as 20 times that of the engines in the leader's aircraft.

In a less extreme example, consider three similar pumps operating in parallel

* Source: Soares, C. M., LCA course notes, 2000.

TABLE L-1 Hypothetical* Sample Format for Component Allowable Lives

Part	Allowed Cycles	Values of Constants
LPC		$c_1 = 1 = c_2$ $k = 0.3$ $p = 3$ (Constants provided by the OEM)
St. 1 disk	6,000	
St. 1 spacer	5,000	
St. 2 disk	10,000	
St. 2 spacer	4,500	
And so on		
LP coupling shafts	40,000 and 20,000	
LP turbine		
Shaft	8,000	
St. 1 disk	5,000	
St. 1 spacer	5,000	
And so on		
HPC		
St. 1 disk	8,500	
St. 1 spacer	4,800	
St. 2 disk	8,000	
St. 2 spacer	7,500	
And so on		
HP coupling shafts	10,000 and 15,000	
HP turbine		
Shaft	15,000	
Disk	5,500	

* Imaginary values chosen for illustrative purposes. These data values are normally proprietary to the individual manufacturers.

in a plant. One of them always seems to have alignment and balance problems, although they all operate at nominally the same process conditions. When they are checked, it is found that extremely high piping stress exerts cantilevered load on the problem pump. The vibration, caused by misalignment and imbalance, even though it may not result in catastrophic failure, may result in this pump requiring overhaul before the other two.

Original equipment manufacturers (OEMs) conduct life-cycle analysis (LCA) to track the validity of their designs as a whole and the performance of newly designed modifications.

Life Usage

Life cycle

A *cycle of life* is a unit that defines a measurable unit of life in a piece of turbomachinery. In many cases, with lower turbine inlet-temperature gas turbines, one cycle is defined as an hour. If the gas turbine always ran at base load, this might be a fair assessment. However, if the gas turbine is at peak temperature, overload, or otherwise stressful condition, an hour of running may add up to more used-up life than if the machine were at base load. If one cycle is what is incurred by running for 1 hour at base load, then more demanding operation will "cost" more.

A usage cycle may or may not, depending on the OEM, embody temperature as a parameter necessary for definition. The OEM may define its cycle for a given model, for instance, by maximum and minimum speed values, each as a percentage of maximum speed versus time.

OEMs might conduct LCA at the request of customers who are concerned about premature parts removal. The reason for removal of parts might vary from machine model to machine model or from component to component. (See Table L-1.)

Alogarithms for life-cycle use

An "algorithm to calculate life-cycle use" is just a technical term for a calculation to determine how much useful life of a component has been used up. If this is information that an operator needs to know (in other words, the OEM may or may not track it for you), then the OEM can give all owners of that machinery model a service bulletin (SB) to cover the details they need to know.

Let us consider the following hypothetical (actual OEM service bulletins to describe their algorithms are proprietary) example that describes the information that an OEM would provide in such an SB.

Format of an LCA SB

1. Number and title of the SB.
2. Model number(s) applicability.
3. Statement of why the SB has been introduced: At operator's request/for improved parts life/optimized operational cost/and so forth.
4. Definition of how the OEM defines "cycle" in terms of
   - Cycle type [for instance, "fatigue cycle" if low-cycle fatigue (LCF) is a factor].
   - Operational parameters that define the cycle [for instance, "a speed change from 0 percent gas generator speed to gas generator (GG) reference speed and back to 0 percent].
5. Calculations and corresponding action outlined in detail, i.e., calculation procedures, definition of reference speeds, which will vary for each rotor, and

$$\text{Sample cyclic usage factor}^{*} = \left[\frac{c_1(N_1)^2 - c_2(N_2)^2}{(N_{\text{ref}})^2 - k(N_2)^2} \right]^p$$

where N_{ref} = reference speed for that rotor
N_1 = highest operating speed
N_2 = lowest operating speed
k = stress at reference speed/ultimate tensile stress at design temperature
p = slope of low-cycle fatigue curve
c_1 and c_2 are constants.

The OEM will specify the calculation sequence. A sequence may be laid out for how to represent all operational peaks in one cycle, which peak to apply the calculation to first, and how to proceed to the next step.

When this is complete, the peaks that equate to a life cycle are totaled to give a cyclic usage factor for that operational cycle only. Some machinery comes with life-cycle counters to automatically add up these life cycles, or the counters and the LCA process may be retrofitted.

For a cycle with only one peak (or plateau), Fig. L-1B, the simplification to Fig. L-1A can be made.

For a cycle with more than one peak (or plateau), see Fig. L-2A. This can be simplified to Fig. L-2B.

Calculation of the first peak (specified as the smallest speed change) is shown in Fig. L-2C. Note dotted line. This then makes the curve shown in Fig. L-2D, which then simplifies to the curve given in Fig. L-2E, so the calculation then is (for constants $k = 0.3$, $p = 3$, $c_1 = c_2 = 1$):

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*Note: This is an actual equation used in a comtemporary gas turbine OEM's SB that saved its end users' component lives by factors exceeding 200 percent.

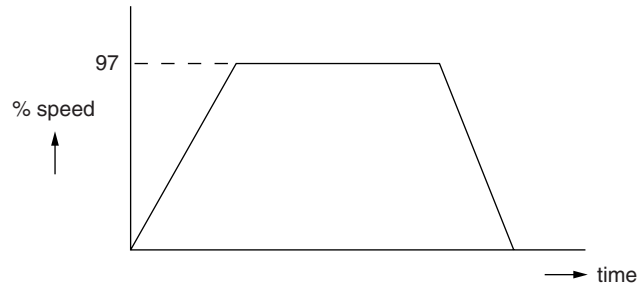


FIG. L-1A Hypothetical curve of percentage (%) speed versus time for a gas turbine.

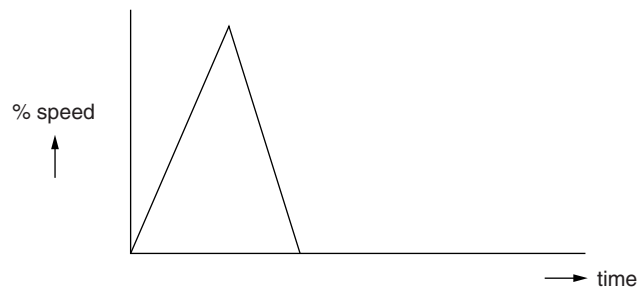


FIG. L-1B Modification of the curve in Fig. L-1A.

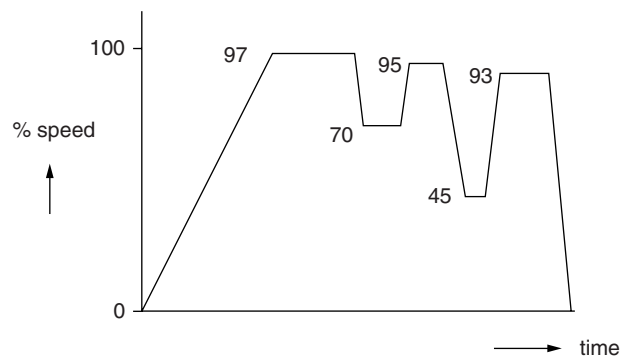


FIG. L-2A A cycle with more than one peak.

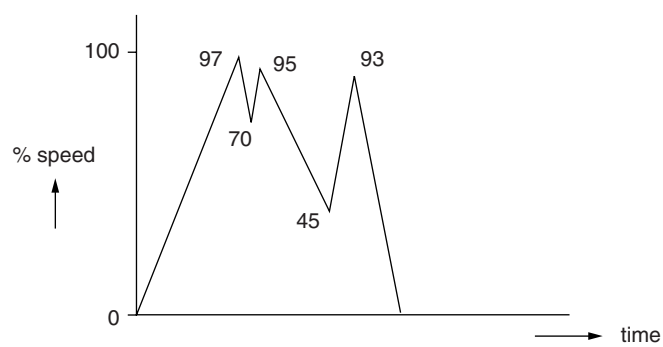


FIG. L-2B Simplification of Fig. L-2A.

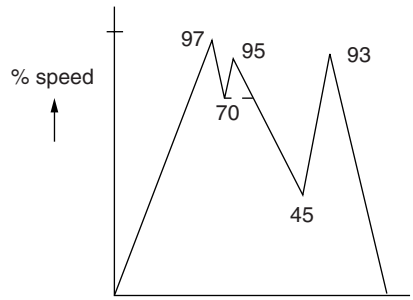


FIG. L-2C Calculation of the first peak.

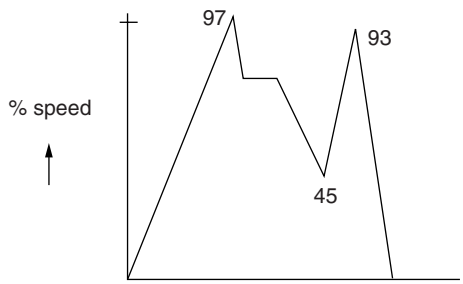


FIG. L-2D Curve after first peak calculation.

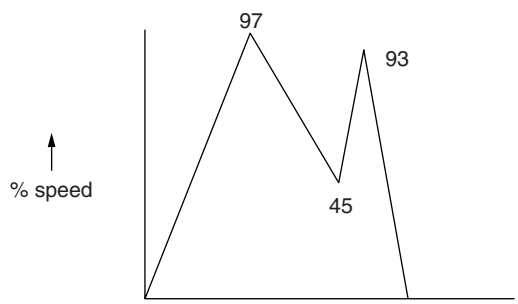


FIG. L-2E Curve simplified after first peak calculation.

$$\text{First peak cyclic usage factor} = \left[\frac{95^2 - 70^2}{100^2 - 0.3(70)^2}\right]^3$$

$$\text{Second peak cyclic usage factor} = \left[\frac{97^2 - 45^2}{100^2 - 0.3(45)^2}\right]^3$$

$$\text{Third peak cyclic usage factor} = \left[\frac{93^2 - 0^2}{100^2 - 0}\right]^3$$

For the overall cycle, add the previous three peaks.

The OEM may also provide graphs that assist with calculation of life-cycle usage. Typically, these would specify curves corresponding to the upper operating speed. The lower operating speed might be along one axis and effective cycle usage on the other axis (see Fig. L-3).

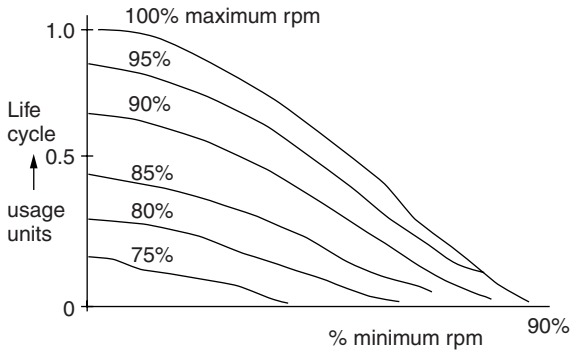


FIG. L-3 Cycle usage curves valid for specific constants.

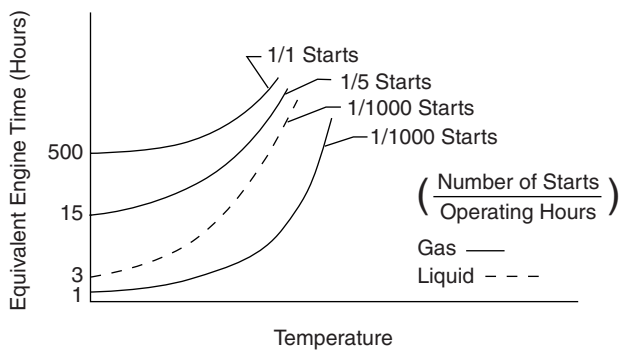


FIG. L-4 Equivalent engine time in the combustor section.

The OEMs would also provide tables of lives for each of the individual components involved. The component with the lowest number of allowed cyclic lives will determine when the unit has to be removed so that component can be changed. The others, however, may be left in until it is economically practical to remove them.

Note that OEMs whose LCA counting uses time and speed, but not temperature, are likely to be quite confident in the cooling capability of their design.

Life-cycle usage in specific applications

Life cycles are always used up faster if the application requires many stops and starts. This is indicated in Fig. L-4. The type of fuel also plays a major role in life-cycle usage, as we will see in the next section. Also, the type of turbine involved affects how many cycles or hours are used up in a normal start. Depending on the algorithms used, these figures will vary. One study estimated that the hours (at base-load conditions) used up, with a stop and start of a large industrial gas turbine (about 20,000 hp), was about 500 h. The equivalent for one of about 3000 hp was 5 h. See also Table L-2 for representation of an independent study of LC usage on an industrial gas turbine. (The algorithms used were not revealed, as is often the case with independent studies.)

Example case history 1. A gas turbine's compressor rotor developed blade tensile stresses while in operation that would quickly lead to failure. The OEM specified that a glass-bead peening process needed to be conducted all over the surface of the

TABLE L-2 Operation and Maintenance Life of an Industrial Turbine

Type and Number Load, Fuel and Starts	Type Inspection (Hours of Operation)			Expected Life (Replacement) (Hours of Operation)		
	Service[f]	Minor[g]	Major[h]	Combination Liners	First-Stage Nozzle	First-Stage Buckets
Base[a,b] natural	+	+	+	+	+	+
Natural gas, 1/1,000	4,500	9,000	28,000	30,000	60,000	100,000
Natural gas, 1/10	2,500	4,000	13,000	7,500	42,000	72,000
Distillate oil, 1/1,000	3,500	7,000	22,000	22,000	45,000	72,000
Distillate oil, 1/10	1,500	3,000	10,000	6,000	35,000	48,000
Residual, 1/1,000	2,000	4,000	5,000	3,500	20,000	28,000
Residual, 1/10	650	1,650	2,300			
System peaking[c,d]						
Natural gas, 1/10	3,000	5,000	13,000	7,500	34,000	
Natural gas, 1/5	1,000	3,000	10,000	3,800	28,000	
Distillate, 1/10	800	2,000	8,000			
Distillate, 1/5	400	1,000	7,000			
Turbine peaking[c,e]						
Natural gas, 1/5	800	4,000	12,000	2,000	12,000	
Natural gas, 1/1	200	1,000	3,000	400	9,000	
Distillate, 1/5	300	2,000	6,000			
Distillate, 1/1	100	800	2,000			

[a] 1/5 = one start per 5 operating hours.
[b] Base = normal maximum continuous load.
[c] No residual usage due to low load factor and high capital cost.
[d] System peaking = normal maximum load of short duration and daily starts.
[e] Turbine peaking = extra load resulting from operating temperature 50 to 100°F above base temperature for short durations.
[f] Service = inspection combustion parts, required downtime approximately 24 h.
[g] Minor = inspection of combustion plus turbine parts, required downtime approximately 80 h.
[h] Major = complete inspection and overhaul, required downtime approximately 160 h.
NOTE: Maintenance times are arbitrary and depend on staff availability and training, spare parts and equipment availability, and planning. Boroscope techniques can help reduce downtime.

compressor blades. This peening would insert a layer of compressive stress that would reduce the net effect of the tensile stress and the rotor could be run safely. Past a certain number of cycles, the compressive stress layer would dissipate and the entire compressor rotor would need to be overhauled again.

During one of the turbine's overhauls, the repair facility mistakenly only glass-bead peened one side of the compressor blades. Two engines failed during operation before the error was discovered. It was not known with which turbines the error had been made. The only way to determine this without doubt was to destructively test the compressor blades with X-ray diffraction. The entire fleet of engines that the overhaul shop had handled with that SB included had to be recalled.

Modifications to Operating Parameters

Parameter adjustment for extension of TBOs (time between overhauls)

Example case history 2. A gas turbine's design required that the first-stage turbine blades be removed at a time halfway between the specified times for major overhauls. The entire row had to be replaced, with one-half of an overhaul cycle of life wasted.

The OEM redesigned the blades. With the newer models, the blades were of a different size, but in terms of LCA, the most important change was to change the

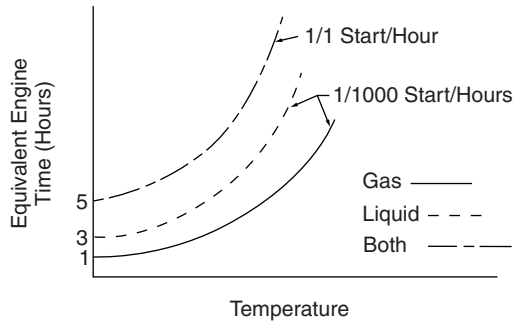


FIG. L-5 Equivalent engine hours—turbine.

material selection, which then affected the actual life cycles used up during operation by changing the stress that the blades saw in operation. The blades were under the stress endurance curve and their life was "limitless and on condition."

Parameter adjustment/machinery choice for different fuel quality

Some gas turbines run on many different kinds of fuel—gas, diesel, and residual fuel. The life of their hot-section components will be very different, depending on the type of fuel they are running. Natural gas is generally a "clean" fuel and causes the least toll in parts life in a gas turbine.

The main parameter adjusted to allow for bad fuel quality is turbine inlet temperature. It is lowered. Frequently, this prompts a choice of a different model of gas turbine or combined cycle (gas turbine/steam turbine) package. Additional features, such as water/steam injection and fuel treatment, may have to be added before life-cycle analysis indicates an economically targeted value for component lives, TBOs, and so forth. See example case history 3.

Figure L-5 shows that liquid fuel causes a higher toll on turbine life, even when an equal number of starts per operating hour is involved.

Design Assessment for Maximum TBOs

OEM design features provide end user options

Example case history 3. Analysis of gas turbine (combined cycle) design features to adapt it to run on residual fuel.

Residual fuel can be hard on the life of machinery parts. The fuel is extremely dirty and requires expensive fuel treatment to be burned, even at reduced turbine inlet temperatures (TITs). Just lowering the topping temperature in the turbine fuel control might not suffice in terms of "hot corrosion" (lowering topping temperature would reduce wear and tear on components). Choice of a different model of gas turbine is what is required to make the situation acceptable.

The ideal turbine for residual fuel application is a relatively low temperature, sturdy, preferably cast, simple design that then results in minimal maintenance. The former ABB GT13D (and its close cousin, the GT11N2), for instance, worked in applications such as the Shunde power plant, Quangdong province, China. (Note that ABB stopped making the GT13D—50 cycles—and replaced it with a GT11N2—60 cycles—and a gearbox. This arrangement costs about 2 percent in efficiency, but consolidates the OME's spares stocking requirements. Note also that ABB is part of the Alstom Corporation as of 2000.) Far greater turbine sophistication with respect to alloys and turbine inlet temperatures would be self-defeating. China, which has residual fuel, needs to run on as cheap a fuel as possible with maximum efficiency and time between overhauls.

Residual oil as a fuel is a potentially difficult prospect with inadequate design features. Corrosion, plugging, and fouling could occur quickly in certain circumstances. Higher firing temperature turbines require complex blade cooling, expensive superalloys, and substantial derating. The -13D has integrally cast blade and vane cooling passages, with relatively simple geometry (versus an aerofoil with laser-produced cooling passages) and large-flow cross sections. This provides better resistance against plugging.

The single-piece welded rotor supported by two bearings is a simple, less vibration-prone design. No through bolts are used, which is another useful maintenance feature.

This design has only one silo combustor and a solid cast design. It has one large bore-fuel nozzle, which helps avoid clogging and erosion. No air atomization is required, which means no compression airstream is required.

The nature of the burner design means that water injection is required. Typically, water injection is 1.3 times the fuel flow rate (maximum 10.5 L/s). Water injection adds 9 to 10 MW of power.

No flow divider is required in this design, so no temperature imbalance is observed in that section. This also helps cut down on maintenance and life-cycle costs of the components in this section.

The generator is driven from the cold end, which means turbine exhaust-end inspections are easier. All bearings are accessible without disassembly and no elbow conduits are required. As the generator is air cooled, no hydrogen system or hazards have to be allowed for. The cooling loop is closed and maintenance free.

The boiler, a vertical-assisted circulation, single-pressure design type, has a preheating loop. It delivers 44 kg/s of 37.5-bar steam at 475°C. Sodium phosphate (Na_2SO_4) is used for anticorrosion measures in steam treatment.

Sodium (Na), sulfur (S), and vanadium (V) content in the fuel are the major problems. Sodium is removed by mixing preheated fuel with water and demulsifier and then centrifuging. Potassium (K) impurities are removed in the same manner and at the same time as the sodium down to 0.5 ppm total (for both Na and K). The sulfur left in the fuel becomes SO_x upon combustion.

Magnesium additives combine with the vanadium to form salts that deposit onto the blade surfaces. When the turbine is shut down, the salts fall off with the drop in temperature. The remaining salts are washed off with plain water. Typically, the wash is done every 100 operating hours for heavy oil. If gas or diesel fuel are used, no wash is required.

[The limit for magnesium addition is 1105°C, as at 1120°C, magnesium oxide (MgO) solidifies to the extent it can only be chiseled off, and V_2O_5 (vanadium oxide), with its low melting point, corrodes. (Both MgO and V_2O_5 are formed from the safe additive compound after 1120°C.) The turbine inlet temperature of such units is typically maintained at 990°C.]

When the gas turbines are started, diesel fuel is used until synchronous speed and then heavy fuel is used. This helps prevent clogging. The turbines are run for 5 min on diesel when shutting down. Again, this prevents clogged nozzles and ignition problems.

The -11N2 can also handle the same rough fuel as the -13D. Peak metal temperatures, internal metallurgy, and fuel-treatment requirements are all quite similar. The single-burner design for this model can get NO_x down to 42 ppm with water injection. An EV silo combustor (several fuel nozzles) option is available if the end user has gas or diesel fuel. NO_x can then be reduced to 15 ppm when at base load on natural gas.

The preceding operational description indicates that fuel choice radically affects the required daily maintenance, overhaul, time between overhauls, and therefore effective life cycles of components. Undoubtedly, fuel choice can radically affect life-cycle assessment results.

However, cheap fuel more than offsets the capital expenditure required for fuel treatment and additives, washing the fuel, and other costs. This cost savings increases with the power capacity of a plant. Using a difference in residual oil and diesel prices of $50/ton, a 300-MW facility typically could save $22 million at 0.5 capacity factor and $36 million at 0.75 capacity factor.

Typical savings of $264 million and $432 million, respectively, are indicated over the life of the plant. It is likely that these figures could increase, as there is a global glut of residual oil.

Risk and weighting factors for design features on various OEM models

When faced with an array of factors embodied in different models that affect LCA, it is advisable to do a risk and weighting analysis. The main steps of this process are as follows.

1. List all your options. Let us say the task is to choose a model of gas turbine that will give you maximum component lives and TBOs. The options would be model numbers of gas turbines. Draw columns representing these options. Subdivide each of these columns into two, for the weighting (priority) and risk (probability).
2. List all the factors that are important to you. These may include:
   a. Maximize component lives
   b. Optimum TBOs
   c. Fuel cost savings
   d. Fuel consumption savings
   e. Avoiding fuel-treatment system requirement
   f. Avoiding water-injection system requirement
   g. Comply with/allow for:
      i. Current environmental regulations
      ii. Future environmental regulation (CO_2 tax)
   h. Minimize expertise/training level required from operators
   i. Maximize efficiency (may be linked with *c*)
   j. Design that is easy to overhaul (aimed at making the overhaul process local and/or in-house) and so forth

   Obviously, some of these factors will work in opposition to each other. List the factors at right angles to the columns in 1.
3. Give a weight (importance) to the factors in item 2. (Use a descending scale of 1 to 10.)
4. Estimate likelihood of these factors occurring, in the right column for each choice. (Use a scale of 1 to 10.)
5. Multiply the weight and the risk. Make the resulting quantity either positive or negative, depending on whether the item favors choice of that option or not.
6. Sum the products in item 5 and arrive at your selection.

Gas-Turbine Deterioration and Diagnosis

Performance recovery

After some hours of operation, the gas turbine(s) in a train may be observed to deteriorate in terms of mechanical condition and/or performance (flow, pressure) deterioration. The deterioration is detrimental to component lives and should be corrected for in the short and long term. Generally, faults with these two aspects of operation can be diagnosed with condition monitoring (vibration analysis included) and performance analysis (PA), respectively. *See* a discussion of vibration

analysis under Condition Monitoring earlier in this book. Performance analysis will be dealt with below.

Metallurgical recovery, refurbishment (*see* Metallurgy)

Steam-Turbine Deterioration and Diagnosis

Performance recovery

After some hours of operation, the gas turbine(s) in a train may be observed to deteriorate in terms of mechanical condition and/or performance (flow, pressure) deterioration. The deterioration is detrimental to component lives and should be corrected for in the short and long term. Generally, faults with these two aspects of operation can be diagnosed with condition monitoring (vibration analysis included) and performance analysis (PA), respectively. *See* a discussion of vibration analysis under Condition Monitoring earlier in this book. Performance analysis will be dealt with below.

Metallurgical recovery, refurbishment (*see* Metallurgy)

Performance Analysis (PA) with Gas Path Parameters

Basically, PA is a check of the fluid path of a piece of turbomachinery (gas turbine, compressor, and so forth) to determine that it is doing what it was designed and bought to do. To perform this check, the pressures, temperatures, and flow along the machinery's gas path may be used to compare its performance curves of pressure versus flow, power developed/required versus flow or other performance parameters, efficiency versus power developed/required, and so forth. The readings may then be computed for the operator to either deduce what corrective action is required or read off the required action on an automated system display.

Performance analysis systems occur in three main formats:

1. Systems that have a calculation module based on actual instrumentation readings. Such systems may cost from about $15,000 up for the first unit, with discounts for additional units of the same model.
2. Systems that have a calculation module based on actual instrumentation readings, as well as a calculation module that uses predictive flow formulas theory and compares the two for better quality information and indicates which instrumentation may be malfunctioning. Such systems may cost from about $60,000 up for the first unit, with discounts for additional units of the same model.
3. Systems that perform the functions of item 2 and are part of an overall on-line (frequently real-time as well) health-monitoring system that incorporates vibration monitoring and other monitoring. Such a system uses artificial intelligence techniques to attempt automated problem solving. Such a system is generally extremely expensive. Typically, these systems cost from about $100,000 per machine train (no discounts for similar trains sometimes) to $250,000 per train.

They may frequently also be "overkill" for an application. Typically, they take about 600 readings/signals as inputs to arrive at their conclusions. Frequently, their specifications also demand requirements that eliminate the competition. Their manufacturers may claim their system's use absolves the operator of using his or her own system knowledge or reasoning by providing "ready" answers. Sometimes this works for certain problems if they were programmed into the artificial intelligence logic. The customers may get the best value for their money from comprehensive vibration analysis and a system that incorporates the capabilities of the system in category 2.

In this section we will discuss a generic system in category 2.* This generally provides the best economic return in terms of dollars saved/dollars of capital cost for a PA system and therefore is one of the most important items in this course.

Consider the following:

- Aims (goals) of a PA system
- Cost, operational benefits, and return on investment (ROI) of a PA system
- How such a system works, so you can compare it against other options and ask manufacturers the right questions
- Advantages and additional applications of PA systems
- Use of a PA system to extend TBOs and reduce repair costs

Aims of a PA system:

1. Check detailed performance for different modules
2. Predict performance through the load range
3. Check machine in normal operation, not just when it has just been cleaned or newly overhauled
4. Check performance of combined cycle (CC) or other complex cycles
5. Self-check and help troubleshoot
6. System must be interactive and simple to run
7. Check cost for different configurations
8. Check cost differential for different operational conditions

When successful, a PA system enhances:

1. Predictive maintenance
2. Reduced performance test cost
3. Improved operations cost prediction
4. Improved system efficiency

In a commercially available PA system, which will be used for illustrative purposes, while setting up reference data, the following parameters were varied:

1. Barometric pressure
2. Ambient temperature
3. Axial compressor flow degradation
4. Axial compressor efficiency degradation
5. Axial turbine efficiency degradation
6. Anti-icing system operation

The diagnostic module input data are (see Fig. L-6):

1. Barometric pressure
2. Ambient dry-bulb temperature
3. Ambient wet-bulb temperature
4. Fuel consumption
5. Compressor delivery pressure

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* Source: For this PA system type and the included example case histories 4 and 5, the information source is Liburdi Engineering, Canada. Material adapted with permission.

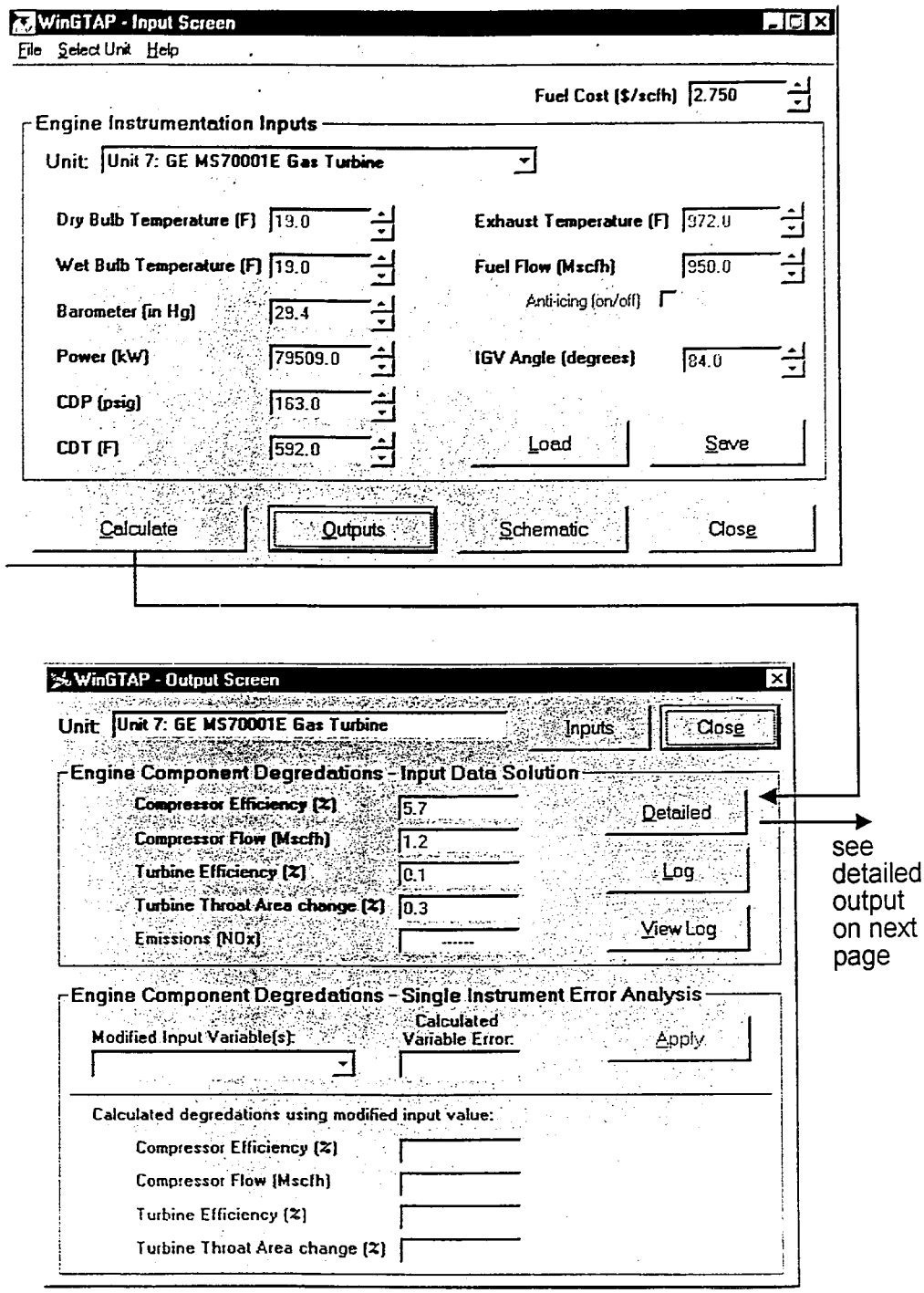


FIG. L-6 Screen displays from PA system (GE Frame 7 engine). (Source: Liburdi Engineering.)

6. Compressor delivery temperature
7. Gross power
8. Exhaust gas temperature
9. Inlet guide vane settings

Any measurement errors are usually with items 4, 5, 7, and 8.

Gas path analysis steps (from the preceding instrument data) are:

1. Calculate compressor inlet flow
2. Calculate TIT from turbine swallowing capacity
3. Calculate turbine efficiency degradation
4. Calculate compressor efficiency
5. Calculate clean compressor flow and efficiency, then calculate flow and efficiency degradation

Experience indicates that the following ranges usually apply:

1. For compressor flow degradation, <0 to >8 percent
2. For compressor efficiency degradation, <0 to >6 percent
3. For turbine efficiency degradation, <0 to >3 percent

An appropriately designed program will calculate items 1 to 3.

In this PA system's predictive module, predictive values were calculated (fuel flows are interpolated).

Fuel consumption was calculated for all combinations of:

1. 3 barometric pressures
2. 5 ambient temperatures
3. 8 degradation combinations
4. 5 to 11 power levels, which gives 600 to 1320 data points

A second set is required if you have an anti-icing system.

Predictive programs are useful for CC, cogeneration, and complex cycles.

This set of grid values can be included in automatic, on-line, or real-time systems.

A practical use of modeling is in the sale of product (gas, power) versus fuel consumption.

When considering PA systems, look for the following features that offer considerable advantages over others:

- Analysis is aerodynamic, not just a heat balance with the machine represented by a black box
- Predicts stage efficiency, pressure ratio, exit angle, gas properties, and swallowing capacity
- Subroutine handles the following losses: profile, incidence, trailing edge thickness, tip clearance, shock, end wall, and secondary
- Cooling air, flame flare, and similar flow interruptions treated as blockages to help accurate modeling

Advantages and uses of having these features:

- Application in sales based on fuel consumption measured more accurately than with normal fuel flowmeters (see Fig. L-7)
- System can be set up to determine optimum combination in a machinery train for a certain load level
- Cooling-flow modifications can be analyzed and then made to prevent problems such as disc cracking
- Similarly, blade cooling modifications can be made

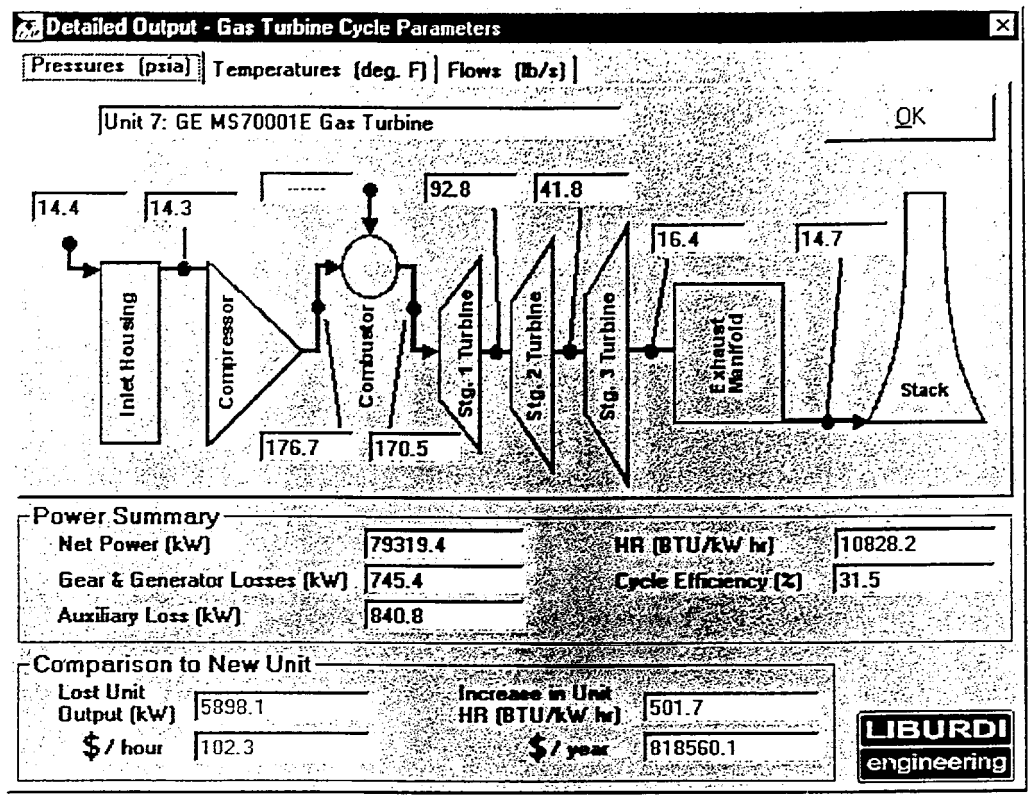


FIG. L-7 Screen display in working PA system (GE Frame 7). (Source: Liburdi Engineering.)

- Analyzing steam- and water-injection effects (reduced maintenance, power increases, NO_x reduction)
- Analysis of gasifiers (GT operated to drive the compressor only) to supply pressurized gas

In one case a W191 was used to provide 1,000,000 lb/h at 10 psig back pressure and below 1000°F. Being able to analyze each stage helped this development of modified running of a gas turbine and aided predictive emissions monitoring.

Note that it should be simple to correct for data inputs that can cause errors due to factors such as:

1. Compressor delivery temperature being affected by heat transfer in the combustor. Solution: Calibrate each engine.
2. Inlet and exhaust pressure loss errors. Solution: Measure periodically.
3. Fuel composition. Solution: Get accurate samples.
4. Modifications [e.g., inlet guide vanes (IGVs)] are added. Solution: Model into solution.
5. Atmospheric condition errors. Solution: Do not forget barometer and relative humidity readings daily (if not incorporated in real time into program).

Example case history 4. Extending TBOs of gas turbines by preventing premature component failure in a GE Frame 5 (see Figs. L-8 to L-10).

Consider that turbine disc lives in some industrial GTs are limited to <100,000 h. This is due to high-temperature creep cracks that are formed when the engine is operated at full load. This, in turn, causes costly repairs and also

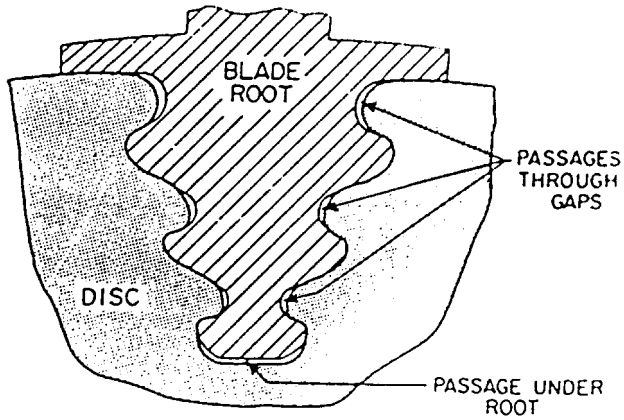


FIG. L-8 Typical blade root-disc serration configuration. (Source: Liburdi Engineering.)

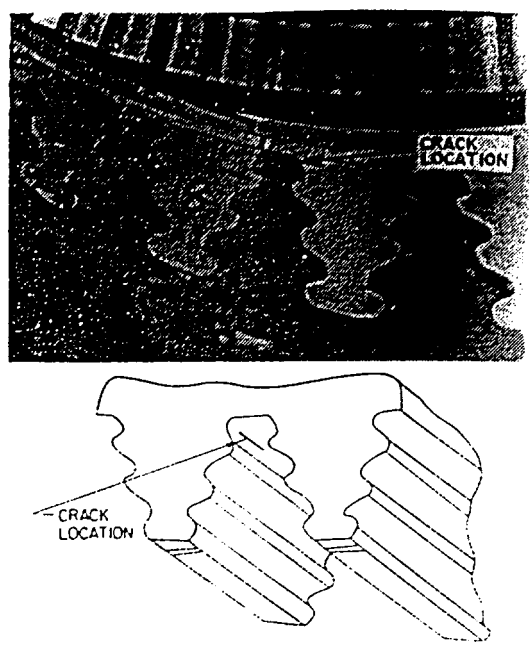


FIG. L-9 The cracks generally start inside on the bottom radius and are very difficult to detect in inspection. (Source: Liburdi Engineering.)

disastrous failures. This can be avoided and disc lives extended to >100,000 h by using performance-monitoring software to analyze changes to the disc cooling.

Note that before this was done, changing the disc material was tried but this did not work. The cracks persisted.

The cracks were at the bottom of the fir tree and difficult to see. Note the following details from the figures:

- Where cracks occurred
- Cracks along grain boundaries
- Root-disc gap configuration
- Compressor air and air hot-gas air paths are located at each disc root. Hot air accumulates where it shouldn't.

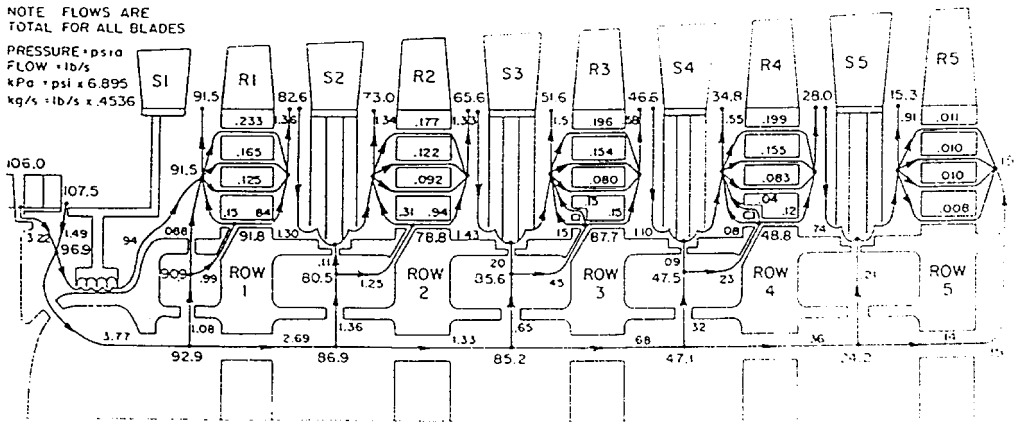


FIG. L-10 "As found" turbine disc cooling flows. (Source: Liburdi Engineering.)

TABLE L-3 Compare Measured and Predicted Values of Engine Parameters (Source: Liburdi Engineering)

	Predicted	Actual
Increase combustor temperature	6.7–7.2	6.7
Increase exhaust temperature	2.2–3.9	1.1
Decrease compressor exit pressure, lb/in^2 (kPa)	0.9 (6.2)	0.7 (4.8)
Increase in fuel flow, lb/s (kg/s)	0.011–0.019 (0.005–0.009)	0.016 (0.007)

- Solution: Feed cool air through diaphragms. No rotating components affected, just the diaphragms. Rows 3 and 4 had compressor delivery air, row 5 had intermediate stage compressor air from bleed valve.
- Diagram of cooling air distribution

(See Table L-3.) Net effect on performance: negligible

- 70-hp (52-kW) increase
- 0.05 decrease in thermal efficiency
- 12°F (7°C) increase in TIT
- 1°C (2°F) increase in exhaust temperature
- 0.75-lb/in^2 (5.2 kPa) decrease in combustor shell

Example case history 5. Power augmentation for a gas turbine in cogeneration service using steam injection. Operation of this system works best when:

- Steam is injected only when a certain power is reached.
- All excess steam is injected and then the control system is allowed to vary IGVs and fuel flow.
- Keep steam lines hot with a small amount of condensate even when steam is not running.

Summary: 30 percent more power is possible when injecting steam equivalent to 7.5 percent of compressor inlet flow. *Note:* NO_x levels are down from 83 to 12 ppm.

Vibration Analysis and Its Role in Life Usage (*see also* Condition Monitoring)

Vibration is a key factor in how long a machine component lasts. The extent to which vibration occurs, and its cause, can be measured by vibration analysis. This is covered in the section on Condition Monitoring.

Note, however, that vibration analysis and performance analysis may be linked in many instances. For example, a cracked combustion liner results in a change in TIT and PA calculations. As the cracked metal disturbs the airflow and is set into a vibration mode of its own, vibration sensors pick up indication of the cracked liner. Depending on the accuracy of the vibration probes, the sensors may pick up the problem before monitoring of gas path parameters.

Vibration analysis is the best detector of problems with components not directly in contact with the gas path, such as bearings, accessory drives, and so forth.

Experienced engineers can do what an expert system does, i.e., arrive at diagnosis of a problem by using indicators from the vibration analysis probes and transducers that are monitoring the gas path.

Example case history 6. The following observations on a compressor could confirm the existence of fouling in the compressor.

Vibration: Rises
PA system data: P_2/P_1 drops, T_2/T_1 rises, compressor efficiency drops
Corrective action: The compressor is washed, and performance recovery is monitored.

For a compressor in surge:

Vibration: Fluctuates, often wildly
PA system data: P_2/P_1 varies, T_2/T_1 does not change, compressor efficiency drops
Other data: Bleed chamber pressure fluctuates, temperature differential across the bearing may be observed to increase, bearing pressure will rise

However, the vibration and the PA system data would be enough to diagnose the high probability of surge.

Example case history 7. For a damaged compressor blade:

Vibration: Rises
PA system data: P_2/P_1 drops, T_2/T_1 rises, compressor efficiency drops
Other data: Bleed chamber pressure fluctuates

Again, the vibration and the PA system data would be enough to diagnose the high probability of surge.

For a compressor bearing failure:

Vibration: Rises
PA system data: No change
Other data: Temperature differential across the bearing rises, bearing pressure drops, bleed chamber pressure stays constant

Note that just the vibration reading should be enough to detect incipient bearing failure or bearing failure, even though not supported (even though not negated) by PA data.

These four cases help illustrate that vibration readings and PA analysis should solve most serious problems. Whether or not the other data back up these two systems, it is not essential to these diagnoses. Very often, marketers of expensive expert systems will try to insist these additional data are vital. While the data may be useful for specific problems, they may not be worth the extra initial capital outlay, as well as cost of operator/engineer training data and/or consultants' fees to interpret the data. (As an example, the fee for consultants to interpret data turned

out by the expert system installed on the Canadian Air Force's small F-18 fleet's F404 engines was about $1 million in 1987. Bear in mind that the expert system could be called justifiable on a critical flight engine, despite triple redundancy in its control systems.)

Codes and Specifications

Specifications for PA systems and intelligent expert on-line systems, real time or otherwise, are as plentiful as the number of system designers/manufacturers. The more expensive they are, the more they are likely to be complex, with an intentional tendency to exclude competition.

Codes for enclosures, such as control panels, computers, controls, valves, and so forth, are unchanged from the codes specified in API, ASME, and so forth, for specifications with respect to safety considerations. *See* Some Commonly Used Specifications, Codes, Standards, and Texts.

Operational Optimization Audits

Audits are conducted to assess the efficiency and validity of a plant, a process or any part thereof at a time during the life of that unit. Audits can result in major, expensive modifications that have a good ROI, such as PA systems. When PA systems are retrofit, this is often the result of an audit, broad or limited in scope.

The word audit carries with it the connotation of time unwillingly but dutifully spent on a necessary evil. The audit team and those who provide them with information expect boredom, witch hunting, paper trails, and, worst of all, lost revenue time. The latter factor may not be the case, depending on the circumstances. With careful planning, the time can be used to optimize design, maintenance, and operational conditions to maximize profit margins. Stricter environmental conditions sometimes make an audit a requirement, and, in some cases, suspended operations. The time should be viewed as an opportunity, as environmentally prompted design changes may herald other significant maintenance time or operational efficiency gains.

There are two kinds of audit teams: internal (in-house) and external. On occasion, the team consists of both of these groups. The audit team is trained to look for areas of material breakdown, safety hazards that have arisen as a consequence of deterioration, and items that require change because they fall under recently enacted legislation.

Note that for circumstances where operational conditions are changing, for instance in a combined oil and gas field where relative volumes of gas, oil, and seawater, as well as molecular weights are changing, the two audit types may occur simultaneously if retrofit, optimization, or redesign become an issue.

Preparation for an audit

1. Collect the data.
   a. Sources include maintenance and production management information systems (MIS), automated and manual, current relevant legislation, and relevant labor contracts.

      Comprehensive MIS can help track recurring items that indicate required specification, design, or maintenance practice changes, such as wear plates instead of wear rings, an additional vibration probe–monitoring position, and additional fluid moved through a seal buffer system. Legislation can dictate abandonment of long used cleaning fluids and procedures and redesign of the exhaust system off a plasma spraying booth. Labor contracts, particularly in a union environment, can dictate similar changes.

      When external changes, legal, labor, or otherwise, dictate a major change

in procedure and/or operating and maintenance procedures, an audit should be considered to cover the scope of all affected systems.

b. Maintenance and production personnels' "must have" and "nice to have" lists and equipment literature.

The status of these items changes through the life of a facility. Where wear rings might have sufficed in abrasive service, changes in process flow content may make wear plates necessary.

An audit, then, is something personnel should plan for and collect data for continuously between audits.

c. Latest updates of relevant standards and practices.

d. Format of paperwork to be used.

e. Description of relevant repair procedures, contractor lists, and spare parts brokers if relevant. Questions asked here should include:

- What is the expected remnant life of the production field in question?
- What are the OEM's service intentions with respect to the models used in production?
- What are spares inventories?
- What are inventories of official scrap of critical components?
- Do new repair technologies make salvage of previously scrapped components possible?
- What impact do the answers to these questions have on the profitable life of the existing plant? On the profitability of planned expansions?
- On the design of planned expansions?
- On the choice of OEMs and system design for planned expansions?

f. Quotes on retrofit procedures and installations.

Contractors should also have indicated their completion times for retrofits for minimum impact on shutdown times. Consider penalty clauses, cost plus clauses, and other relevant expense items.

2. Planning process.
   a. Get updates of all information in step 1.
   b. Identify departments that should have audit input.
   c. Identify the extent of input required from different departments.
   d. For each department identify primary and secondary contacts.
   e. Formulate a time-line program. Work backward from the required completion date of the audit.
   f. Review the time-line schedule with the team.
   g. Decide on the interface of audit/regular operations/ongoing maintenance/shutdown.
   h. Finalize the time line. (Time line should be flexible at all times.)
   i. Identify and build special tooling/gauges/instrumentation.
   j. Identify any special heavy lifts required. Arrange all details of safety equipment required. Relevant questions may include:
      - For critical rotor balancing procedures, will specific arbors make fewer operations possible?
      - Will tolerance tightening on specific balance tooling decrease rotor imbalance and increase TBOs?
      - Will digital versus analog readouts affect operational efficiency? TBO?
   k. Identify the tolerance changes required by specific applications.
   l. Identify and collate information learned from previous equipment failures.

Recommendations for conducting the audit

1. Using the information collected during the preparation phases, formulate the checklists to be used during the audit. The lists are only to be used as guides, however, as totally unforeseen circumstances might come to light.

2. Members of the audit team should include representatives from all departments that may be affected by its outcome.
3. Provide audit team members with appropriate training conducted by an external objective party. This party should work in concert with plant personnel and OEMs but not be focused on any specific party's interests.
4. An objective party, preferably the trainer in item 3, should be present during the audit and during analysis of its findings.
5. Arrange for relevant photographic records to be made and filed during the audit for future analysis.

Summary

For life-cycle analysis to be truly successful, it needs to be linked with everyday operations and maintenance at a plant, as well as with periodic audit and shutdown activities. The amount of equipment and instrumentation used for LCA should be tailored strictly to just what is necessary. A great many expensive "bells and whistles" (features) may be unnecessary and just produce mounds of additional data that the customer has to manage.

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Liquid Eliminators (*see* Separators)

Liquid Natural Gas (LNG)*

An LNG processing system requires filters and other appropriate accessories to maintain appropriate delivery properties. A basic system is shown in Fig. L-11. This is an area where constant research is being conducted to minimize vessel size and weight. Computational fluid dynamics (CFD) and specialized probes assist in this research and can, when necessary, also be used in operational functions to avoid plant shutdowns (see Figs. L-12 through L-14).

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* Source: Peerless, USA.

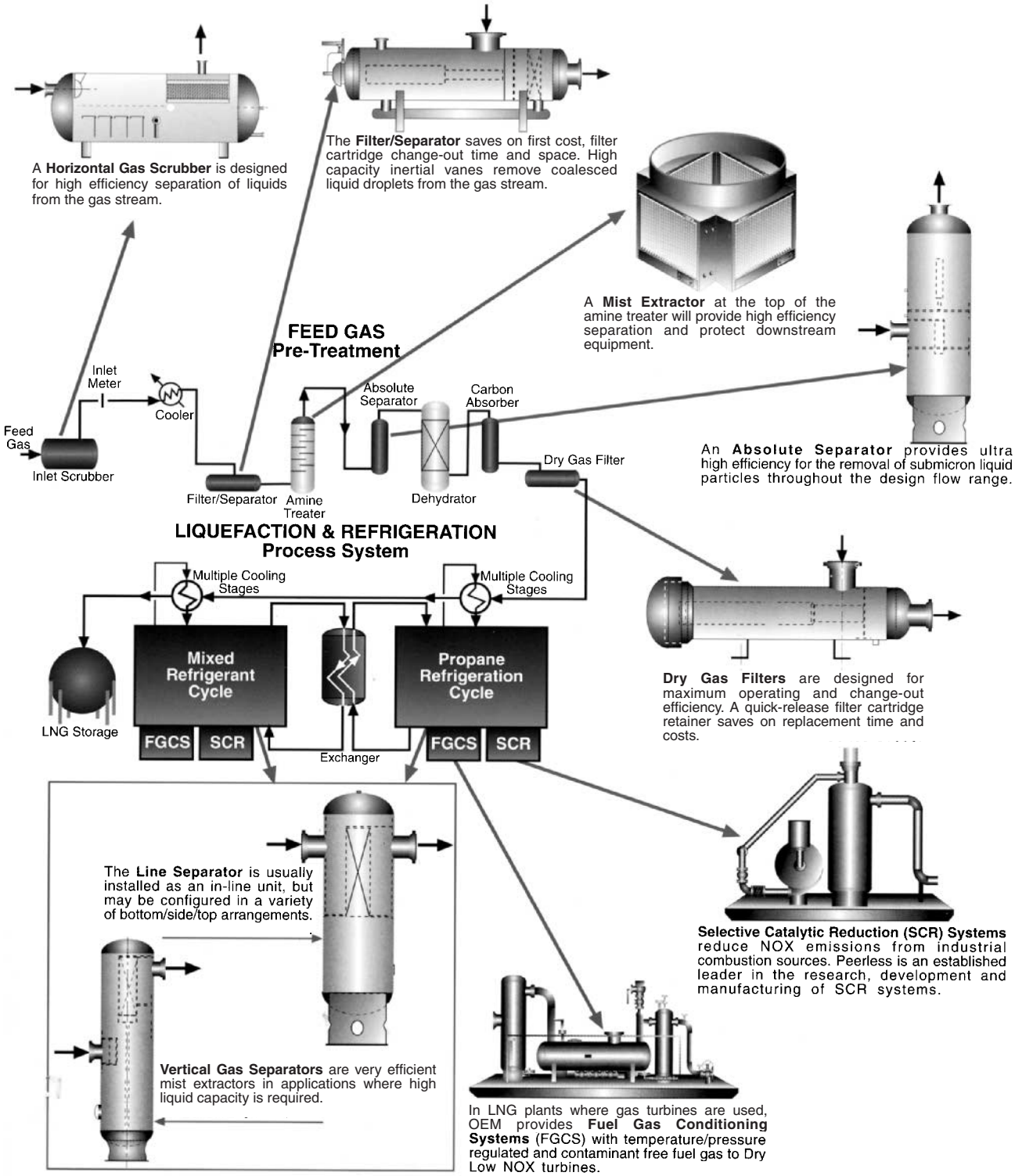


FIG. L-11 An LNG feed, liquefaction, and refrigeration process system. (Source: Peerless.)

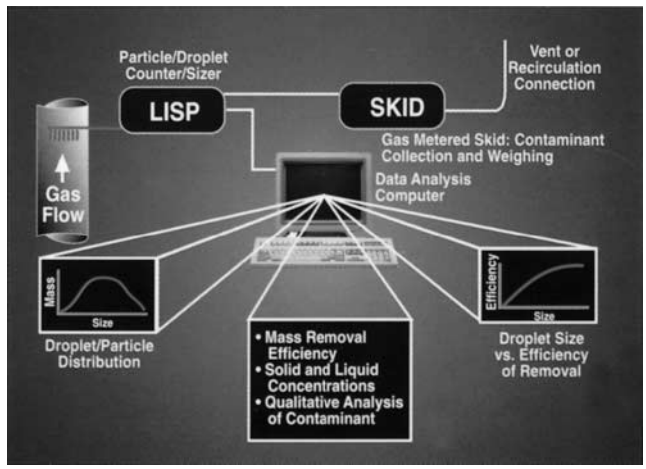


FIG. L-12 Diagram of the Laser Isokinetic Sampling Probe (LISP^{SM}) field test setup and field analysis equipment. (Source: Peerless.)

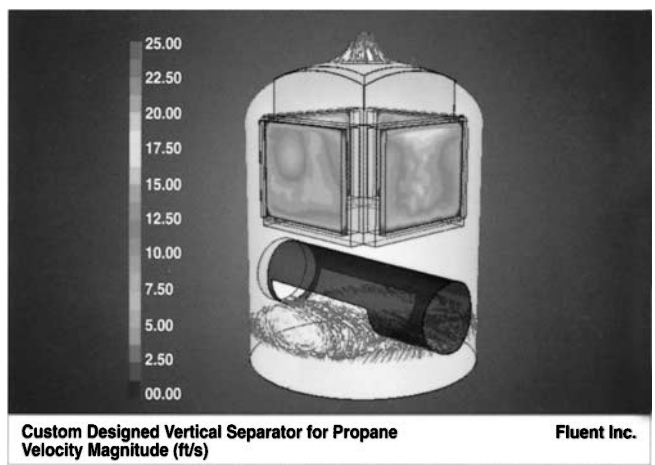


FIG. L-13 Proprietary Sizing^{TM} reduces the vessel by several sizes. Computational fluid dynamics technology contributes to the application solution and ensures all design specifications are met. (Source: Peerless.)

Typical Liquefied Natural Gas Process

Computational fluid dynamics (CFD)

Sophisticated computer models help to reduce the size of separator vessels and ensure that liquid/vapor separation is achieved to specification. The CFD flow model pictured in Fig L-13 depicts the final design of a vertical gas separator for an LNG facility. This graphic provides the engineer with visual confirmation of gas flow paths and that the separator face velocities meet established design criteria.

CFD models use actual vapor properties such as those for propane, ethane, or any of the various mixed refrigerants to determine separation performance and capacity.

In-line testing without plant shutdown

A new field sampling tool for pressurized gas streams, the Laser Isokinetic Sampling Probe (LISP^{SM}) was developed, custom-designed, and built to specifications. It collects

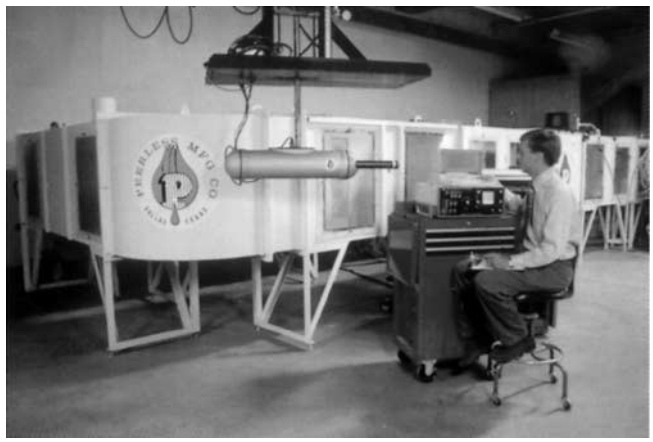


FIG. L-14 An R&D lab is equipped with a computerized forward scattering spectrometer probe (FSSP). The FSSP uses precision optics and a laser to measure liquid droplets down to submicron diameters. This FSSP is being inspected before being placed into the wind tunnel. (Source: Peerless.)

and weighs entrained liquids and solids both up- and downstream of separators or filters at very high system operating pressures.

Thus, samples can be taken of liquids and solids in their pressurized state. And because of the high degree of sensitivity demanded by the LISP, meticulous measurements can be made of particles as small as 0.3 microns in diameter. The result is the most accurate and reliable pressurized, in-line, field sampling of LNG processes without a plant shutdown.

Lubrication*

Lubrication is primarily concerned with reducing resistance between two surfaces moving with relative motion. Any substance introduced on or between the surfaces to change the resistance due to friction is called a *lubricant*. In addition to reducing friction, a lubricant removes excess heat, cleans microscopic wear particles from surfaces, coats surfaces to prevent rust and corrosion, and seals closures to prevent dust and moisture from entering.

The choice of the proper lubricant not only is important to manufacturers in order to enable them to meet their guarantees for performance and reliability but is, of course, of the utmost importance to users of the equipment in keeping their maintenance costs to a minimum and safeguarding machinery against abnormal wear, corrosion, and the effects of contamination. When choosing a lubricant, conditions such as operating speed, load conditions, method of sealing, temperature range, moisture condition, bearing design, and quantity of lubricant all affect the final choice.

It is generally recognized that a specification giving only physical and chemical properties does not guarantee satisfactory performance of any particular lubricant. Manufacturers and users, therefore, must rely on the experience, integrity, and

* Source: Demag Delaval, USA.

reputation of the lubricant supplier and on the record of satisfactory past performance of the particular type of lubricant offered for a given purpose.

The lubricant should be a first-grade branded product that has previously been used and proved to be satisfactory for the continuous lubrication of similar equipment in the same service. Such experience should have proved the lubricant to be satisfactory, particularly with respect to foaming, rusting, sludging, and separation for water and other impurities.

The brand of lubricant decided upon should be continued in use and should not be changed without compelling reason.

Lubrication Methods

Either splash lubrication or forced-feed oil lubrication is commonly used for rotating machinery such as turbines, pumps, compressors, reduction gears, and worm gears. Splash lubrication is used for relatively slow-speed machinery, while high-speed machinery always requires forced-feed lubrication.

The usual form of splash lubrication employs oil rings. In this arrangement a loose ring rides freely on the journal and dips into a sump in the bearing bracket containing oil. The ring rotates because of its contact with the journal, but at a slower speed. The oil adheres to the ring until it reaches the top of the journal, when it flows onto the shaft.

Ring oiling for small machines is used predominantly when the additional cost of a pumping system cannot be justified. The system enjoys the advantage of self-containment, needing no external motivation for its performance. Cooling coils are sometimes added when the sump temperature may become excessive.

The fully forced, or direct-pressure, system, in which the oil is forced into the bearing under pressure, is used in the majority of large circulation systems. Force feeding increases considerably the flow of lubricant to the bearing, thereby removing the heat generated by the bearing. This system is most reliable in high-speed operations with considerable load. (See Figs. L-15 and L-16.)

Grease lubrication is principally used for ball bearings and roller bearings since the housing design and maintenance are simpler than for oil lubrication. As compared with an oil system, there are virtually no leakage problems and no need for a circulation system.

The data in Table L-4 give desirable viscosities and other specifications for oils. The data in Table L-5 give grease recommendations for various applications.

Oil Characteristics

A lubricating oil should be a petroleum oil of high quality having guaranteed uniformity, high lubricating qualities, and adequate protection against rust and oxidation. It should be free from acids, alkalies, asphaltum, pitch, soap, resin, and water. The oil must not contain any solid matter or materials that will injure the oil itself or the parts it contacts or impair its lubricating properties. Lubricating oil should not foam, form permanent emulsions, oxidize rapidly, or form sludge. It may contain additives or inhibitors if their use supplements but does not adversely affect the desirable properties and characteristics of an oil.

Horsepower losses, bearing exit temperatures, and oil-film thicknesses decrease with lower viscosity values and increase with higher viscosity values.

When cold starting is important or a product has ring-oiled bearings, a lubricating oil with a high viscosity index should be used. A high viscosity index means that the rate of change of viscosity of an oil with change of temperature is small.

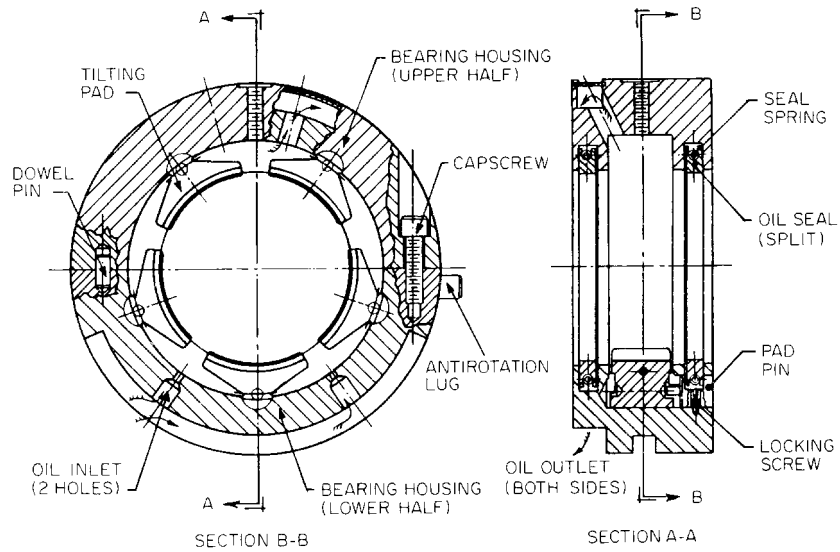


FIG. L-15 Typical tilting-pad bearing. (Source: Demag Delaval.)

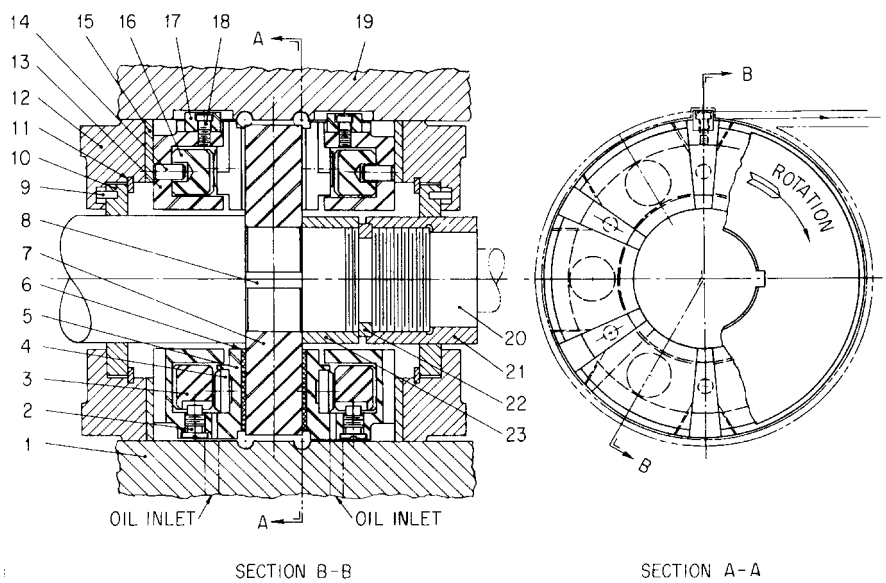


FIG. L-16 Section of tilting-pad thrust bearing. (1) Bearing bracket. (2) Leveling-plate set-screw. (3) Upper leveling plate. (4) Shoe support. (5) Shoe. (6) Shoe babbitt (4, 5, and 6 assembled as a unit). (7) Collar. (8) Key. (9) Pin. (10) Oil guard. (11) Snap ring. (12) Thrust-bearing ring. (13) Base ring (in halves). (14) Leveling-plate dowel. (15) Shim. (16) Lower leveling plate. (17) Base-ring key. (18) Base-ring key screw. (19) Bearing-bracket cap. (20) Shaft. (21) Outer check nut. (22) Retaining ring. (23) Inner check nut. (Source: Demag Delaval.)

Grease Characteristics

Greases should be high-grade, high-temperature lubricants suitable for application by hand, pressure gun, or hand compression cup.

Greases should remain in the solid state at operating temperatures. Grease components should not separate on standing or when heated below their dropping

TABLE L-4 Oil Selection (Source: Demag Delaval)

Product					Type of Oil	Viscosity, SSU ASTM D88			Oil Temperature, °F	
						100°F	130°F	210°F		
						Minimum Maximum	Minimum Maximum	Minimum Maximum	Minimum Operating	Normal to Bearings
Marine propulsion units: turbine-driven					Turbine*	490 625	220 270	62 minimum	90	110 130
Ship's service turbine-generator sets marine auxiliaries: direct or gear drive					Turbine	375 525	180 230	54 minimum	90	110 130
Marine propulsion units: diesel-driven					EP, R&O	630 770	270 320	69 minimum	90	110 130
Centrifugal pumps	Direct drive	Ring-oiled bearings	With water cooling		Turbine	250 350	120 155	47 minimum	. . .	140 160
			Without water cooling	Liquids up to 130°F	Turbine	250 350	120 155	47 minimum	. . .	140 160
				Liquids 131°F and above	Turbine	375 525	180 230	54 minimum	. . . . . .	140 180
			Forced circulation		Turbine	250 350	120 155	47 minimum	. . .	110 120
		Forced feed			Turbine	140 180	85 105	42 minimum	90	110 120
	Gear drive	Forced feed			Turbine	250 350	120 155	47 minimum	90	110 120
Gears	Helical				Turbine	140 180	85 105	42 minimum	90	110 120
	Worm				See Sec. 10, Table P-11					
Turbines	Direct drive	Ring-oiled bearings	With water cooling		Turbine	250 350	120 155	47 minimum	. . .	140 160
			Without water cooling		Turbine	375 525	180 230	54 minimum	. . .	140 180
			Forced circulation		Turbine	250 350	120 155	47 minimum	. . .	110 120
		Forced feed			Turbine	140 180	85 105	42 minimum	90	110 120
	Gear drive	Forced feed			Turbine	250 350	120 155	47 minimum	90	110 120
Centrifugal compressors	Direct drive				Turbine†	140 180	85 105	42 minimum	90	110 120
	Gear drive				Turbine†	140 180	85 105	42 minimum	90	110 120
Turbochargers	Lubricated independently				Turbine	375 525	180 230	54 minimum	. . .	120 160
	Lubricated by engine				See engine manufacturer's specifications	SAE 20 or 30 preferred			. . .	120 160

* Approximately 300 lb/in Ryder gear machine test. † Compressors with oil seals, 190 minimum aniline point.

TABLE L-5 Grease Selection (Source: Demag Delaval)

Component		Type	NLGI No.	Worked Penetration	Drop Point	Corrosion Test
				Minimum ASTM D 217–68	Minimum ASTM D566–76	
Ball bearings, roller bearings, oscillating or sliding plain bearings, sliding pedestal supports*		Sodium, lithium, or sodium-calcium soap base	2	265–295	350	Pass federal test method Standard No. 791 Method 5309.2
Governor valve lifting gear	Steam temperature 600°F maximum	Sodium or lithium soap base	2	265–295	350	Pass federal test method Standard No. 791 Method 5309.2
	Steam temperature 600–825°F	Nonsoap base	1 or 2	265–340	500	Pass federal test method Standard No. 791 Method 5309.2
	Steam temperature over 825°F	Silicone	1 or 2	265–340	520	Pass Military G-23827 A

* An alternative lubricant for sliding-pedestal supports is a mixture of fine graphite and cylinder or turbine oil mixed to a paste consistency.

point, the temperature at which grease changes from a semisolid to a liquid state. They also should not separate under the action of centrifugal force.

Greases should resist oxidation and must not gum, harden, or decompose. They must not contain dirt, fillers, abrasive matter, excessive moisture, free acid, or free lime.

Oil Maintenance

The lubricating system must be kept clean and free from impurities at all times. The accumulation of impurities will cause lubricant failure and damage to the equipment.

Provision should be made for maximum protection against rust during idle periods. The main lubricating system should be operated at intervals to remove condensation from metal surfaces and coat these surfaces with a protective layer of lubricant. This should be done daily when the variation in day and night temperatures is great and weekly when the variation in day and night temperatures is small. In addition, a unit idle for an extended period of time should, if possible, be operated from time to time at the reduced speeds specified under normal starting procedures.

The use of a suitable oil purifier is recommended. Since some purifiers can alter the properties of lubricating oils, especially inhibited oils, the manufacturer should be consulted before the purifier is selected.

Grease Maintenance

Grease housings should be relubricated routinely when the grease in service is unable to satisfy lubrication demands. The housing should be completely flushed

and filled with new grease and any excess worked out before replacing the drain plug. Care should be taken not to overfill the housings, as this will result in a breakdown of the grease to fluid consistency and overheat the bearings.

In some cases, small additions of fresh grease to the housing are sufficient for proper lubrication. When this procedure is followed, the housing should be completely cleaned and new grease added during each major overhaul.

Lubrication Piping

Oil-feed and oil-drain piping is generally of low-carbon steel. Piping used should be pickled (a procedure of cleaning the internal surfaces). If low-carbon steel piping has not been pickled, the following procedure should be followed:

1. Sandblast pipe along the pipe run.
2. Deburr if necessary.
3. Wash all internal surfaces with a petroleum-base cleaning solvent.
4. Air-blast dry.
5. Visually inspect.
6. If piping is to be stored in house, fog all internal surfaces progressively along the pipe run, through all openings, with an oil-soluble preservative compound.